

Rerun pipeline — student run guide

This script set replaces all three original notebooks. Every number in the paper (`papers/adversarial-mri-detection/`) comes from running it.

Ground rules:

- **Every run is designed to be reproducible and auditable.**
- Every stage verifies the **SHA-256 of each model against** `results/manifest.json` before doing any work — if a model file has been touched, the stage errors out immediately. This is deliberate: a silently retrained model is exactly the bug that invalidated the original results.
- Training scripts **refuse to overwrite an existing model** — you must pass `--force` yourself, understanding that every downstream result becomes invalid.

Environment

```
# Python 3.12 + TF 2.20 (on the GPU machine this lives in WSL2)
python3 -m venv ~/venvs/tf220
~/venvs/tf220/bin/pip install "tensorflow[and-cuda]==2.20.*" scikit-learn
pandas matplotlib pillow
```

- Working directory: `~/ThammasatResearch/` (override with the `TR_BASE` env var)
- Place the dataset at `~/ThammasatResearch/dataset/archive.zip` — its sha256 must be `8828172500...fdb4` (the scripts verify this for you)
- Recommended: `export CUDA_VISIBLE_DEVICES=0` and `export TF_FORCE_GPU_ALLOW_GROWTH=true`

Run order (main pipeline — in this folder)

#	File	What it does	Time on an RTX 3070
0	<code>common.py</code>	Shared helpers: seeding/ determinism, dataset guard, manifest, FGSM (not run directly)	—
1	<code>01_train_classifier.py</code>	Train the MobileNetV2 classifier (seed 42, deterministic) and record its hash/accuracy in the manifest	~2.5 min
2	<code>02_fgsm_sweep.py</code>	FGSM sweep, ϵ 0–8, on the test set	~1.5 min
3	<code>03_detector_v2.py</code>	Train the consistency detector (baseline, calibrated on the full validation set)	~5 min
4	<code>04_final_test_v2.py</code>	Final test of detector v2	~1.5 min
5	<code>03b_detector_v2_calibrated.py</code>	Train detector v2b: val_A (model selection) / val_B (threshold) split	~5 min
6	<code>04b_final_test_v2b.py</code>	Final test of v2b — demonstrates the FPR transfer failure	~1.5 min
7	<code>05_deployment_calibration.py</code>	Deployment calibration: Testing → 400 calib / 1,200 eval, threshold from clean images only	~2.5 min
8	<code>06_pgd_eval.py</code>	PGD (K=10/40) against the detector at the pinned threshold	~20 min

9	<code>06b_intersection_analysis.py</code>	FGSM/PGD comparison on the intersection of success sets; persists per-image scores to <code>attack_scores.npz</code>	~20 min
—	<code>make_figures.py</code> (in <code>papers/.../exp1/</code>)	Generates the paper's Fig 2–3 from the CSVs	seconds

Run them in order:

```
cd ~/ThammasatResearch/scripts/rerun
~/venvs/tf220/bin/python 01_train_classifier.py
```

(then each subsequent file the same way)

Audit scripts (dataset integrity)

⚠ **These are not standalone:** they must run from inside this folder (they import the stage modules via `import_module("03_detector_v2")` etc.), and several require artifacts in `results/` produced by earlier stages (`attack_scores.npz`, `clean_scores.npz`, `text_banner_scan.csv`, `pixel_leakage_scan.csv`, `manifest.json`) — they assert immediately if anything is missing (intentional, fail-closed).

File	What it does
<code>scan_text_banners.py</code>	Scans all 7,200 images for burned-in banners/text → found the glioma collection shift (1.2% vs 13.3%)

<code>check_aug_leakage.py</code>	⚠ This file's results were REFUTED — never trust its numbers: 64-bit perceptual hashes false-alarm catastrophically on MRI slices (it reported 93.9% near-identical, which is wrong). Kept unmodified as an audit-trail lesson; real duplication numbers come from <code>pixel_leakage_scan.py</code> only
<code>pixel_leakage_scan.py</code>	The correct duplication detector (pixel-level) → found 100/400 meningioma test images duplicated from Training
<code>verify_duplicates.py</code>	Confirms candidate duplicate pairs at full resolution
<code>exclusion_and_shortcut.py</code>	Detection rates under exclusions + the shortcut check (glioma collapses on banner images: 15.1%)
<code>clean_fpr_exclusion.py</code>	Clean FPR under exclusions + calibration sensitivity
<code>make_example_figure.py</code>	Generates Fig 1 (per-class MRI examples, exemplars screened to be clean)
<code>diag_score_shift.py</code>	Diagnostic of the clean-score distribution shift (validation vs test)

Expected checkpoint numbers (verify against the paper)

- Clean test accuracy **0.81875 (1,310/1,600)** — if you get anything else, your environment differs; stop and investigate rather than pushing on.
- Stage 7: threshold 0.3612, eval FPR **0.0983**
- Stages 8–9: PGD detection within ~1 point of FGSM on the intersection

Full provenance: `papers/adversarial-mri-detection/exp1/PROVENANCE.md`
and `papers/adversarial-mri-detection/review-synthesis.md`